

Objective

Passionate experimental nuclear physicist building a career in nuclear physics, aiming to advance the field through comprehensive, end-to-end research. Committed to driving forward the development of nuclear technology and applications, and fostering innovation in experimental approaches and methodologies.

Summary of Qualifications

Research and development experience in accelerator mass spectrometry, nuclear, and space physics with high-impact publications in *Physics Letters B* and *Physical Review Letters*, along with presentations and seminars delivered at international conferences and laboratories.

Extensive experience with operation of heavy ion accelerators, radiation detectors, associated data collection systems, data analysis, and radiation safety, including heavy-ion, gamma-ray, electron, positron, and neutron detection.

Author of high impact policy paper on implementing Australia's nuclear submarine program working with the Australian Strategic Policy Institute.

Programming and computer expertise in digital data acquisition, scientific computation, and data analysis in Python, C++, and Fortran

Led the first measurements of $E0$ transitions in Cr isotopes and ^{24}Mg using the ANU Super-e spectrometer. ^{24}Mg results published in PLB.

Redesigned the NIM/VME timing electronics for the Miel Si(Li) detector array, improving timing resolution by an order of magnitude to ≈ 10 ns, enabling the first-ever observation of an $E0$ from a super-deformed band-head state in ^{40}Ca , published in PRL

Education

2017 – 2022
Mar Mar

Doctor of Philosophy in Nuclear Physics, *Australian National University*, Canberra
Awarded 18 March 2022

Dissertation: An investigation into nuclear shape coexistence through high precision conversion electron and electron-positron pair spectroscopy

2016 – 2016
Feb Dec

Bachelor of Science (Honours), *Australian National University*, Canberra
First class

Thesis: The search of double internal conversion in ^{137m}Ba

2013 – 2016
Mar May

Bachelor of Science, Physics and Mathematics, *University of Auckland*, Auckland
A+ average

Employment

- 2025 – Jan **AMS Specialist Programme Support Officer**, *Australian National University*, Canberra
Accelerator mass spectrometry of Cl-36, development of gas ionization detectors, analogue and digital data acquisition systems, along with analytical chemistry and sample preparation, external user support, and business development and outreach.
- 2023 – 2024 Mar Feb **Postdoctoral Research Scholar**, *University of Kentucky*, Lexington
 γ -ray and fast-neutron spectroscopy using HPGe and liquid scintillator detectors. Ran the particle accelerator and laboratory, built and maintained the electronics and data acquisition system, assisted visiting users and supervised students, drafted publications and presented work at conferences and laboratories.
- 2022 – 2023 Aug Feb **ANU Research Officer**, *Australian National University*, Canberra
Physicist at the Department of Nuclear Physics at the ANU to develop and commission the new space radiation testing beam line. Drafted documentation, manuals, user guide, and advertising material for the new beamline.
- 2022 – 2022 Jan Sept **Medical Physics Associate**, *Canberra Hospital*, Canberra
Part of the Radiation Oncology medical physics team. Performed dosimetry quality assurance, linac commissioning measurements, radiation detector testing, and wrote reports.
- 2019 – 2022 Apr Mar **Physics Laboratory Demonstrator**, *Australian National University*, Canberra
Designed, developed, and delivered physics laboratory experiment for PHYS3711 - Noise and Measurement, and for the Fundamentals of Nuclear Science course for the Masters of Nuclear Science program.
- 2018 – 2021 Aug Mar **Physics Outreach**, *Australian National University*, Canberra
Explained physics and gave back to the community through tours, demonstrations and talks. Part of the RSPE team for Science in ACTION, ANU Open Day, the National Youth Science Forum, ConocoPhillips Science experience programme, and demonstrations for school groups.
- 2015 – 2015 Feb Jul **Research Assistant**, *University of Auckland*, Auckland
Research assistant to Dr Graham M. Donovan to research and publish work on lung airway dynamics. This resulted in a publication on the behaviour of airway smooth muscle and airway dynamics.
- 2014 – 2015 Jul Nov **Physics Laboratory Demonstrator**, *University of Auckland*, Auckland
Lead first year physics laboratory groups and tutorials for: PHYSICS 140 - Digital Fundamentals; PHYSICS 109 - Planets, Stars and Galaxies; and PHYSICS 160 - Physics for the Life Sciences.

Highlighted Publications

- 2024 Sept **Identification of very large $E0$ transition strengths in ^{52}Cr : Shape coexistence and nuclear structure**, *J. T. H. Dowie, et al.*, *Physics Letters B* (in preparation)
- 2024 Aug **Level lifetimes in ^{52}Cr from DSAM following inelastic neutron scattering: implications for shape coexistence and $E0$ strengths**, *J. T. H. Dowie, et al.*, *EPJ Web of Conferences* (Accepted)
- 2021 Dec **Implementing Australia's nuclear submarine program**, *Andrew Nicholls, Jackson Dowie and Marcus Hellyer*, *The Strategist*, <https://www.aspistrategist.org.au/implementing-australias-nuclear-submarine-program/>
- 2021 Apr **Electric monopole transition from the superdeformed band in ^{40}Ca** , *E. Ideguchi, T. Kibédi, J. T. H. Dowie, et al.*, *Physical Review Letters* **128**, 252501 (2022), <https://doi.org/10.1103/PhysRevLett.128.252501>
- 2020 Jul **Evidence for shape coexistence and superdeformation in ^{24}Mg** , *J. T. H. Dowie, et al.*, *Physics Letters B* **811**, 135855 (2020), <https://doi.org/10.1016/j.physletb.2020.135855>
- 2020 Apr **Evidence for shape coexistence in ^{52}Cr through conversion-electron and pair-conversion spectroscopy**, *J. T. H. Dowie, et al.*, *EPJ Web of Conferences* **232**, 04004 (2020), <https://doi.org/10.1051/epjconf/202023204004>

- 2020 **Table of electronic factors for $E0$ electron and electron-positron pair conversion transitions**, *J. T. H. Dowie, T. Kibédi, T. K. Eriksen, A. E. Stuchbery*, Atomic Data and Nuclear Data Tables **131**, 101283 (2020), <https://doi.org/10.1016/j.adt.2019.06.002>
Jan
- 2016 **Airway compliance and dynamics explain the apparent discrepancy in length adaptation between intact airways and smooth muscle strips**, *Jackson Dowie, Thomas K. Ansell, Peter B. Noble, Graham M. Donovan*, Respiratory Physiology & Neurobiology, <https://doi.org/10.1016/j.resp.2015.09.006>
Jan

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Conference Presentations

- 2023 **Structure of $^{50,52,53,54}\text{Cr}$ via $(n, n'\gamma)$ and DSAM: implications for shape coexistence and $E0$ strengths**, *Talk*
Jul
Capture Gamma-ray Symposium 17 (CGS17), Grenoble, France
- 2019 **Investigating shape coexistence in $^{50,52,54}\text{Cr}$ with $E0$ transitions and pair-conversion spectroscopy**, *Talk*
Oct
American Physical Society, Division of Nuclear Physics Fall Meeting 2019, Arlington, Virginia, U.S.A.
- 2019 **Exploring Shape Coexistence between doubly magic ^{40}Ca and ^{56}Ni through pair-conversion spectroscopy**, *Talk*
Sept
Heavy Ion Accelerator Symposium 2019, Canberra, Australia
- 2019 **Exploring Shape Coexistence between doubly magic ^{40}Ca and ^{56}Ni through pair-conversion spectroscopy**, *Talk*
Jul
International Nuclear Physics Conference 2019, Glasgow, Scotland
- 2019 **Shapeshifting Nuclei and Electric Monopole Transitions**, *Poster*
Jan
Global Young Scientists Summit 2019, Singapore
- 2018 **The evolution of shape coexistence between doubly magic Ca-40 and Ni-56 through pair-conversion spectroscopy**, *Talk*
Dec
2018 Australian Institute of Physics Congress, Perth, Australia
- 2016 **The search for Double Internal Conversion in ^{137m}Ba** , *Talk*
Dec
2016 Australian Institute of Physics Congress, Brisbane, Australia

Invited Seminars

- 2024 **Shape coexistence via conversion electron and electron-positron pair spectroscopy**
Jan
Brookhaven National Laboratory (BNL), USA
- 2023 **Shape coexistence at the heart of $N=28$: $E0$ transitions and inelastic neutron scattering**
Aug
Lawrence Livermore National Laboratory (LLNL), USA
- 2023 **Electron (and positron) spectroscopy with the superconducting magnetic spectrometer (Super-e)**
May
Oak Ridge National Laboratory (ORNL), USA
- 2019 **The evolution of shape coexistence between doubly magic ^{40}Ca and ^{56}Ni through pair-conversion spectroscopy**
Jan
University of Oslo, Norway

Service and Outreach

- 2025
Sept **HIAS 2025 Editor**, *HIAS 2025*, Canberra
Editor for the proceedings of the Heavy Ion Accelerator Symposium 2025.
- 2019
Oct **Conference Experience for Undergraduates Mentor**, *APS DNP Meeting*, Arlington, VA
Served as a mentor for the Conference Experience for Undergraduates (CEU) program at the APS DNP meeting. Interacted with CEU students during the meeting to help them navigate the conference, be a friendly face, and attend their research posters in order to enhance their conference experience.
- 2019
Jan **Global Young Scientists Summit 2019**, *Singapore*
Represented ANU and RSPE at the Global Young Scientists Summit, 2019 (GYSS) as one of four postgraduate students from the ANU. Presented my research as a poster to Nobel laureates and fellow PhD students from around the world. Represented in GYSS 2019 Highlights video, https://www.youtube.com/watch?v=ou4BQ_xNKDG
- 2018 – 2021
Sept Mar **RSPE HDR Mentor**, *Australian National University*, Canberra
Mentor in the peer mentoring program for higher degree by research students.
- 2017
Jul **HIAF Online Tour: Target Area One video**
Video describing the experimental target area one in the Heavy Ion Accelerator Laboratory at the Australian National University: <https://physics.anu.edu.au/tour/nuclear/?Location=TargetAreaone&Article=TargetAreaOne>
- 2017
Jun **RSPE Profiles: Exciting nuclei with strange shapes**
Profile article about me as a physics student at the Australian National University: https://physics.anu.edu.au/news_events/profile.php?ProfileID=4
- 2017 – 2021
Mar Mar **Nuclear Physics Beer Scholar, XXIII**, *Australian National University*, Canberra
Continued the tradition of the department from 1965 and contributed to the culture of the department by organizing social Friday evenings for all department members.

Awards and Honours

- 2020
Mar **COVID-19 Extension Scholarship (Research)**, *Australian National University*, Canberra
Scholarship awarded by the Australian National University providing an additional stipend to high performing students for postgraduate studies for an additional year due to disruptions from COVID-19 restrictions.
- 2020
Mar **Postgraduate Research Scholarship**, *Australian National University*, Canberra
Scholarship awarded by the Australian government to continue to fund my PhD program and stipend for an additional year.
- 2019
Jun **Robert and Helen Crompton Award**, *Australian National University*, Canberra
Awarded to fund part of my PhD overseas for 3 months making collaborative visits to the University of Kentucky Accelerator Laboratory and TRIUMF, Canada, along with funding my travel to present at the American Physical Society, Division of Nuclear Physics Fall Meeting.
- 2018
Sept **Director's Prize, John Carver Seminar Series**, *Australian National University*, Canberra
Awarded for seminar presented as part of the John Carver Seminar Series 2018 at the Research School of Physics and Engineering, ANU for the talk titled *Shapeshifting Nuclei: how and where to find them*
- 2018
Sept **John and Silva Newton Award**, *Australian National University*, Canberra
Awarded from the HIAF endowment to attend and present my research at the Global Young Scientists Summit 2019 (GYSS 2019) in Singapore.
- 2017
Mar **ANU Supplementary Scholarship**, *Australian National University*, Canberra
Scholarship awarded by the Australian National University providing additional stipend to high performing students for postgraduate studies.

- 2017 **Australian Government Research Training Program Domestic Scholarship**, *Australian National University*, Canberra
 Mar
 Scholarship awarded by the Australian government to fund my PhD program and provide stipend for three years.
- 2016 **Senior Scholar Award**, *University of Auckland*, Auckland
 Jan
 Awarded to best performing student in the Bachelor of Science cohort finishing at the end of 2015 majoring in Physics and Mathematics.
- 2016 **First in Course Awards**, *University of Auckland*, Auckland
 Jan
 Awarded first in course awards for being the highest achieving student in a course in 11 physics and mathematics courses.

Professional Memberships

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| 2019 | American Physical Society (APS) | <i>Member</i> |
| 2018 | Australian Institute of Physics (AIP) | <i>Member</i> |
| 2018 | Australian Nuclear Association (ANA) | <i>Member</i> |

Skills and Experience

Experience in accelerator mass spectrometry, nuclear, medical, space, radiation physics.

Expertise in conversion electron, electron-positron pair spectroscopy, and Electric monopole transitions

Scientific computation and data analysis with C++, Fortran, Python

Experience in Monte Carlo simulation for radiation transport using PENELOPE and GEANT4.

Good communication skills, with experience in teaching, supervision, and writing internal reports, external publications, conference and seminar presentations.

Extensive hands-on experience with light- and heavy-ion accelerators, radiation detectors, signal-processing electronics, and data acquisition systems.

Radiation detector experience: Gas ionization chambers, HPGe, Si(Li), NaI, LaBr, and liquid scintillator detectors

Experience with analogue NIM/VME electronics and XIA/CAEN digital data acquisition.